Name ____ Date ___________ Period _____

Parent Functions IV: b^x and log base b

TEACHER KEY | Algebra II Honors | Unit 1 | Day 7 | TEK A2.2.A

Objective: I can graph exponential and logarithmic parent functions, analyze their attributes, convert between exponential and logarithmic form, and explain why their domains and ranges are exchanged.

Vocabulary

Exponential function — the variable sits in the **exponent**. **Base** — b = 2, 10, or **e ≈ 2.718**.

Logarithm — log base b of a answers: what **exponent** on b gives a? So a logarithm IS an exponent.

Two forms — $b^x = a$ and log base b of a = x say the **same thing** two ways.

Part 1 — Two Tables That Trade Places

x	-2	-1	0	1	2	3
$f(x) = 2^x$	1/4	1/2	1	2	4	8

x	1/4	1/2	1	2	4	8
$g(x) = \log_2(x)$	-2	-1	0	1	2	3

State precisely what happened between the two tables: **the rows swapped - inputs became outputs**

Can 2^x ever be 0 or negative? **no** So what is the domain of $\log_2$? **$(0, \infty)$**

Part 2 — Converting Between Forms

Rewrite each in the other form. A logarithm is just an exponent asked for backwards.

Exponential form	Logarithmic form	Exponential form	Logarithmic form
$2^x = 8$	**$\log_2(8) = 3$**	$10^x = 1000$	**log(1000) = 3**
$5^x = 25$	log base 5 of 25 = 2	**$2^x = 1/8$**	$\log_2(1/8) = -3$
$3^x = 81$	**log base 3 of 81 = 4**	$e^x = 1$	**ln(1) = 0**

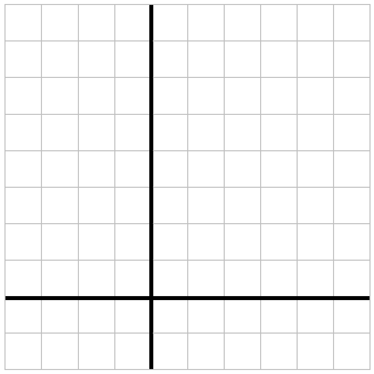
Solve by inspection: $2^x = 32 \rightarrow x =$ **5** $\log_2(64) =$ **6** log(100) = **2**

Why is log base b of 1 always 0, for any base? **any base to the power 0 equals 1**

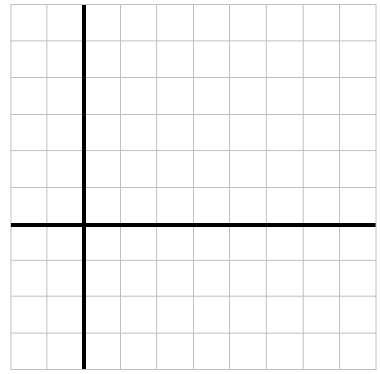
Part 3 — Graph Them Both

Each square is 1 unit. Draw the asymptote dashed first.

A. $f(x) = 2^x$



B. $f(x) = \log_2(x)$



HA **$y = 0$** intercept **(0, 1)**

VA **$x = 0$** intercept **(1, 0)**

Part 4 — Attributes Side by Side

Attribute	**b^x**	**log base b**
Domain	**$(-\infty, \infty)$**	**$(0, \infty)$**
Range	**$(0, \infty)$**	**$(-\infty, \infty)$**
Intercept	**(0, 1)**	**(1, 0)**
Asymptote	**$y = 0$**	**$x = 0$**

Every row is swapped. Explain what relationship between the two functions causes that:

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They are **inverses** — formalized on Day 10.

Part 5 — Going Further

1. Which grows faster past $x = 1$, 2^x or 10^x? Justify with two values.

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2. Every exponential graph b^x passes through one shared point. Name it and explain. **(0, 1), since $b^x = 1$ when $x = 0$**

3. Why is log base b of a negative number undefined?

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No exponent on a positive base **produces a negative result**

Exit Ticket

A function has domain $(0, \infty)$, range all reals, x-intercept (1, 0), and a vertical asymptote at $x = 0$.

Family: **logarithmic** Inverse family: **exponential** Rewrite $\log_2(16) = 4$ in exponential form: **$2^x = 16$ with $x = 4$**